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Pricing Strategies

RELEVANT COSTS FOR PRICING DECISIONS

Objectives

- understand why incremental, avoidable, and forward looking costs are relevant for pricing
- understand two-stage decision making and how fixed and sunk costs are accounted for in this process
- recognize opportunity and relate to pricing decisions.

Summary of Main Points:

Clearly identify the decision!

- The relevant costs may change with the decision.
- Relevant costs for a pricing decision may change throughout an organization. A retail buyer may face different costs than a retail store manager.

Which costs are relevant for pricing decisions?

- Incremental vs. Fixed
- Avoidable vs. Sunk
- Current vs. Historical Watch out for historical costs!

Recognize opportunity costs.

Uncertainty may explain why opportunity costs are seldom recognized in practice. For example, it may be difficult to price some assets (e.g. selling a used machine). It may also be difficult to evaluate the payoff of some strategies.

The appropriate hurdle for a project is not 0 - it is the payoff of the next best project.

Most variable costs are semi-fixed or lumpy.

- How do you practically deal with this issue?
- When is averaging semi-fixed costs a reasonable approximation? Will accounting for semi-fixed costs change my decision?
- Average costs are often used. Questions to ask - What are you averaging over? What is the variance?

When can you ignore small, incidental costs?

- When the ratio of the small costs/margin is small. If margins are thin, small costs matter.

Allocated costs - how do we cover all our costs?

- Two stage problem: Pricing leads to maximizing the contribution, then ask whether this covers fixed costs (Example: Diamond Courier Service).
- Don't let allocated costs affect pricing decisions!

Pricing to cover costs, marking up over cost, setting price to achieve a return on sales or capital investment are all incorrect. Price targets often suggest minimum prices for staying in business, not maximum prices (again 2-stage problem).

Limitations and Assumptions of Economic Model

The basic economic model that we use assumes that a firm faces a cost function, C(Q), and can produce products at a cost that follows this function. The first problem faced by firms is measuring this cost function. Activity based costing and other accounting practices are all aimed at trying to better understand the cost of an item.

Our basic pricing rule sets MR = MC. However, in practice we often have poor information about both marginal revenue and marginal cost. While there is a large gap between theory and practice, the gap is being closed with better measurement systems.

Theory vs. Practice

Opportunity Costs

Perhaps the single biggest gap between theory and practice on the cost side are opportunity costs. These are perhaps the most difficult costs to grasp because they capture potential opportunities. Lack of use becomes most apparent when one looks at new business proposals. Pick up any new business proposal and you will find a set of cash flows and expenditures for the project. The benchmark for the new proposal is almost always "Will this project make money?" This is of course the wrong criterion to use. One really wants to know which project, or set of projects, will generate the most money.

As we consider a complex set of pricing strategies, we wish to apply the same rule. We want to choose the pricing strategy, or set of pricing strategies, that lead to the greatest contribution to the firm. Opportunity costs are captured in the payoff in the next best strategies that we could pursue.

Identify the Decision

Accounting systems report a number that is invariant to the decision context. However, in theory we know that the context of the decision affects the relevant cost. Capacity utilization, inventory levels, etc. potentially affect the relevant cost. However, the current levels of these variables may not be incorporated in a cost reporting system.

All costs are lumpy
Perhaps the biggest difference between theory and practice is that nearly all costs are lumpy in practice. For example, a restaurant may serve a steak dinner with a baked potato. What is the marginal cost of the steak or the potato? It is more than likely that the restaurant purchased case of steaks and a bag of potatoes and that the average cost is used as the marginal cost of the steak or potato. If there is limited spoilage and these items can be stored and used over time, average costs are probably close approximations of marginal costs.

More generally, purchasing is generally done in large quantities (i.e. cases, truckloads, barrels, etc.) but our decision rule applies to individual units. That is, we need to know the marginal cost of the last unit. This is where managerial judgement is needed to discern whether the average cost is the appropriate cost to use. For long-run decisions with large quantities of production, average costs may be appropriate. When the lumpiness of purchasing and production have a significant effect on profitability, averaging may lead to poor decisions.

"I manage the big picture - incidental costs don't matter!"
Yes, one should focus on the big issues and set smaller issues aside. When looking at cost, one often looks for cost drivers. That is, what are the most significant components of our cost structure? It is assumed that attention to these cost drivers will lead to the most efficient means to manage overall costs. Incidental or small costs will factor into the bottom line, but don't need to be managed as closely.

This is obviously correct, but may often lead to the erroneous conclusion that incidental costs do not need to be considered for pricing decisions. It is assumed that because the ratio of incidental costs to total costs is small that they can be ignored in pricing decisions. There is no disputing that the ratio is small. However, this is the wrong reason to ignore these costs. A correct reason to ignore these costs is that the costs are a small fraction of the contribution margin. This is the ratio that matters.

Consider the following example. A firm sells CD players at $100 retail. Cost drivers for the firm are materials and labor which are typically $90. We tend to ignore handling costs that add $3 to total costs. The $1 in small costs are indeed a fraction of total costs - $1/$90 = 1%. However, the ratio of $1 to the margin of $10 is a whopping 10%. Without these small costs in our pricing decision, our margins are $10. With the small costs factored in, our margins shrink to $9 or 10%.

If our margins on CD players are $100, the small cost only has a 1% impact on profits. Hence, when margins are large relative to small costs, one can more liberally ignore small costs. When margins are small, ignoring small costs is dangerous.

Allocated Costs

Surveys of leading manufacturing firms indicate that fully loaded is used in practice for the vast majority of firms. In fact few firms use only marginal cost in pricing decisions. On the surface, it appears there is a large gap between theory and practice. But perhaps this is not the case.

In developing a business plan, managers look at the NPV of a project. As such, allocated costs are incorporated into the business plan - and for the most part they should. Many of these costs are variable in the long-run, and for long-run planning should be considered. The problem here is one of survey measurement. If firms don't believe they will recover all their costs, they should not pursue a project (NPV < 0). Hence, if you ask a manager, "Did you consider all costs or just marginal costs when setting price?" the question itself is not clear. In the very short-run, marginal costs matter. In the long-run, all costs are variable. Hence, a survey may not accurately reflect what firms do.

Danger of Pricing Rules
Cost based pricing rules are common in practice. For example, historical data at a catalog firm reveal that merchants (pricing manager) typically use a percent markup rule over marginal cost. The danger of such rules is obvious - ignoring demand. However, a closer look at the pricing rules of this firm reveals that the rules are quite sensible. On basic items (socks, underwear, tshirts) and seasonal high-end fashions, the markup rules are quite different. The markup rules vary due to the merchant's beliefs about the relative competitiveness of these products.

The message: pricing rules that fail to incorporate demand are bad, but this does not imply pricing rules are bad in general.

Conclusion
The remainder of this document considers a number of cost examples.

Example 1: Identify the Decision
Your division wants to expand and take over an additional building that is currently empty. The company wants to know whether it should allow your division to expand into the building or lease the building to an outside firm. The accounting value of the building is zero as it has been fully depreciated. We know that the economic cost is greater than zero. Your manager wants to know what costs are relevant to you as a pricing manager?

First, let’s consider some possible answers:

- Opportunity cost of the building: You could sell the building, lease the building, or use the building for a different division.
- Variable costs of operating the building: Maintenance, electricity etc.

While both of these answers are relevant to the true economic cost of the building, the question is ambiguous because no pricing decision was identified. The relevant economic cost may change depending on the decision. Two questions that are more clear:

A building is empty and your boss wants to know whether it is in the best interests of the firm to move into the building. Which costs are relevant for the decision to move into the building?

Which costs are relevant for the pricing decision of your product?

As pricing manager, you first consider what price you would set if you were in the building. The costs relevant to this decision are incremental, avoidable, and forward looking costs. To fully advise your boss, you next need to consider whether this venture is profitable. The costs relevant to this decision are that opportunity cost of the building. If you don't move into the building, what is the next best use of the resource?

Notice that the decision often pins down what is fixed and what is variable. For short-run decisions, many costs are unavoidable. For long-run decisions, all costs become avoidable. Thus, one question to ask is whether this is a temporary price change or a long-run change in strategy.

The bottom line is that one needs to clearly identify the decision before one can assess which costs are relevant.

Example 2: Economic Costs vs. Accounting Costs
Assume that you are about to move into a new building that cost $5 Million to build. The real estate market suddenly collapses and the building is now worth $2.5 Million. Which costs are relevant for the decision to the decision to move into the building vs. lease?

The accounting value is not the appropriate value to use. The appropriate value is the opportunity cost. That is, the next best use of that resource. This may be the market price, the contribution of moving an alternative division into the building, the contribution from leasing, etc. The $5 Million is sunk.

Example 3: Replacement Costs vs. Historical Costs

a) Suppose you lend a can of coffee to a friend. You bought the can on sale for $3 and the current price of a can of coffee is $5. What would you say to your friend if they re-paid you $3? You probably would not want to lend them coffee again ...

Similarly, suppose you paid $5 for a can of coffee. Your thrifty neighbor stops in and asks to borrow the can of coffee for a dinner party tonight. Your neighbor calls the store to get a can of coffee delivered to you the next day. Since coffee is on sale this week, the current price of a can of coffee is $3 and delivery is free. The individual that lent the coffee only wants the coffee replaced. The fact that they paid $5 is irrelevant. For example, when the coffee arrives and you realize it only cost $3 to replace you don't ask your neighbor for $2 extra to cover your original $5 expenditure. What you paid in the past is irrelevant.

b) You are the purchasing agent for Starbuck's coffee. You purchase 6 months of premium coffee at $10 per pound. (The accounting department will treat this cost at $10 per pound.) After you place your order, there is a frost in South America. The price of these same coffee beans is now $15.

You now want to consider a decision to raise your prices. What is the appropriate cost to consider when setting your prices?

If Starbuck's can buy and sell at the market price, the appropriate cost to use is the current price of beans. Why? Consider the following thought experiment. Assuming transaction costs are not significant (and this may be a big assumption), Starbuck's could theoretically sell all of its supply of "$10 beans" at the current market price. They could then repurchase the same inventory of beans at the market price of $15. (Note that this transaction does not require physical movement of the beans if there is a market to buy and sell beans.) With an inventory of $15 beans = market price of beans, they would then use this cost in the pricing decision. The appropriate cost is the current replacement cost of the beans, not their historical cost.

c) Now suppose you are a small grocery store. After a frost in South America, you decide that you want to get out of the premium coffee business and focus on other products.

Some questions to ask:

> Do you have access to a bean market? At what cost? At what price can you sell them on this market?
>
> The market price of the beans is not necessarily the appropriate cost. Remember, the opportunity cost is the next best use of that resource. If you are a large, well connected firm like Starbuck's, the assumption of being able to buy and sell at close to the market price may be reasonable. If you are a small firm, this assumption seems a bit strong.

Example 4: Wines and Sunk Costs
Sherwood Deutsch from Century Liquors went to France in May 1996 to buy wine. This wine won't be bottled and delivered until 1997 or later. However, Woody completed his payments on the wine December 1996.

Fast forward to 2025. The 1996 harvest was fantastic and Woody is nearly sold out of this vintage. He has one remaining case in his cellar. As far as Woody knows, there are no other cases of this vintage available. Against his better judgment, Woody decides to set a price for the wine and sell it at his store. What is the appropriate cost for Woody to consider when setting the price of the wine?

The appropriate cost in 2025 is 0. The price that Woody paid in 1996 is irrelevant to maximizing profits today. Woody should set the price such that he generates as much revenue as possible from the sale of the case of wine. Whether Woody paid \$300 for this case or \$30,000 for the case, if he is going to sell the case he wants to maximize the contribution from the sale. The price paid in 1996 is sunk.

If there is a market outside his store for the wine, this offers Woody the option of selling the wine in his store at a given price or selling it on the outside market. Woody would be foolish to sell the wine at his store at a price lower than the outside market. This just sets a minimum price that Woody should accept, not a maximum price. Again, if he is going to sell the wine he should maximize the contribution which means selling at the highest possible price.

Example 5: Designer Dresses and Sunk Costs

a) It is April, and you are a buyer for Talbot's placing your orders for the Fall season. The high-end fashions will be sold during the fall season at one price. At the end of the season you will give away the unsold dresses to charity.

You will place an order for a specific quantity of dresses. At the time of purchase, you also select a price that you will print on each price tag. What costs are relevant to your pricing and order decision?

Clearly, the wholesale price of the dresses (your marginal cost) is the relevant cost. The buyer will set MR(q)=MC(q) to find q. Then, the buyer will print p(q) on the price tag.

b) It is the start of the Fall season. You own 70 dresses and you paid \$20 per dress to obtain the dresses. At the end of the season, the dresses will have a value of zero. You face a demand curve p = 100-q. If you can only sell at one price during the entire season, what price should you select?

Given that the dresses have already been paid for, this is a sunk cost. The optimal decision at this point is to maximize revenue. You would set MR=MC=0. In this case, MR = 100-2q so q=50 maximizes total revenue. The price is also 50. The contribution from the sale of dresses is 2500. If one wants to account for the cost of the dresses, the net profit is \$2500 - 70*20 = 1100.

Suppose I used the \$20 in my pricing policy. Setting 100-2q = 20, I would select q=40 and p = 60. The contribution from the sale of dresses is 2400. My net profit, after accounting for the cost of the dresses is \$2400 - 70*20 = 1000. Thus I am worse off.

Finally, notice that when I solve for q*=50, I also had at least 50 dresses to sell. If I had fewer than 50 dresses to sell, I would select a higher price. For example, if I had 20 dresses to sell, I would select a price of 80. My contribution is 1600. Higher prices would not be optimal since the contribution decreases as the price increases. (e.g. at p=90, q=10, Contribution=900)

c) Bringing it all together ...

You may be asking yourself, "Is there an inconsistency with (a) and (b)?" In one case, we use the wholesale price in our pricing decision, but once the dresses arrive in the store, we ignore the wholesale price? Let's consider the same example and show that the decision rules are consistent.

Again assume that demand is p = 100 - q, MC = $20. In (a), we set MR=MC so q*=40, p*=60. In (b) we ignore MC and seek to maximize revenue. Our revenue maximizing price and quantity is q*=50 and p*=50. BUT, we only have 40 dresses to sell!! Given that we have 40 dresses to sell, the revenue maximizing price is p*=60, the same as in (a). Thus if there is no uncertainty, we should see 40 dresses priced at $60 whether in (a) or (b).

Example 6: Capacity Constraints and Opportunity Costs
We will often encounter situations where we face capacity constraints. For example, in the Healthy Spring Water example, the trucks may have excess space. How does one think about the excess capacity on a given truck?

As we have said all along, the appropriate cost is the next best use of that resource. If the truck could carry other items to a store (e.g. bottled juice), then one would have to consider this as the opportunity cost of the capacity.

A common example that we all have faced is excess capacity on an airline. Airlines are sometimes full and sometimes have excess capacity. How should one think about the opportunity cost of a seat on an airplane?

Suppose that you can sell seats at two fares: $200 and $50 one way. Suppose a plane travels at full capacity with some $200 passengers and some $50 passengers and there is unlimited demand of $50 seats. Clearly, the airline incurs an opportunity cost of having $50 passengers if there was a customer willing to pay $200 that could not board the flight. The contribution that could have been earned is $150.

Reserving a seat for a $200 passenger who MAY arrive is costly. The expected gain from this seat is the probability that it is filled times the revenue earned on the seat. The airline will allocate the seat if this is greater than the marginal benefit of removing the restriction, which is $50.